Retail Profitability Analysis: Evaluating Discount Impact, Product Performance, and Regional Inefficiencies

1. Executive Summary

This project analyzes retail sales performance to evaluate key drivers of profitability and identify sources of financial inefficiency. While total sales reached $2.27M, overall profit was limited to $282.86K, with 1,808 loss-making orders contributing to -$154.51K in losses.

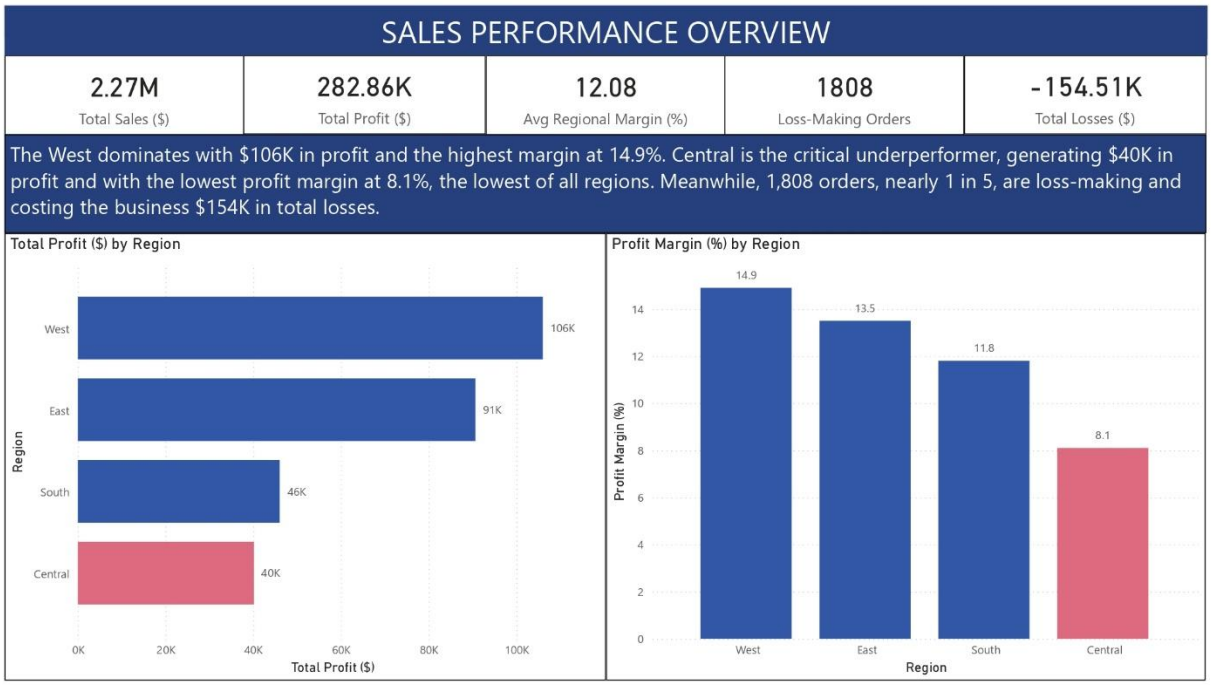
The analysis reveals that profitability is uneven across regions, product categories, and individual products. The Central region underperforms, operating at the lowest margin (8.1%), while excessive discounting significantly reduces profitability. At the product level, several high-selling items generate minimal or negative profit, indicating inefficiencies in pricing and discount strategies.

Overall, the findings highlight structural gaps in pricing, product mix, and regional performance that limit the company's ability to convert revenue into sustainable profit.

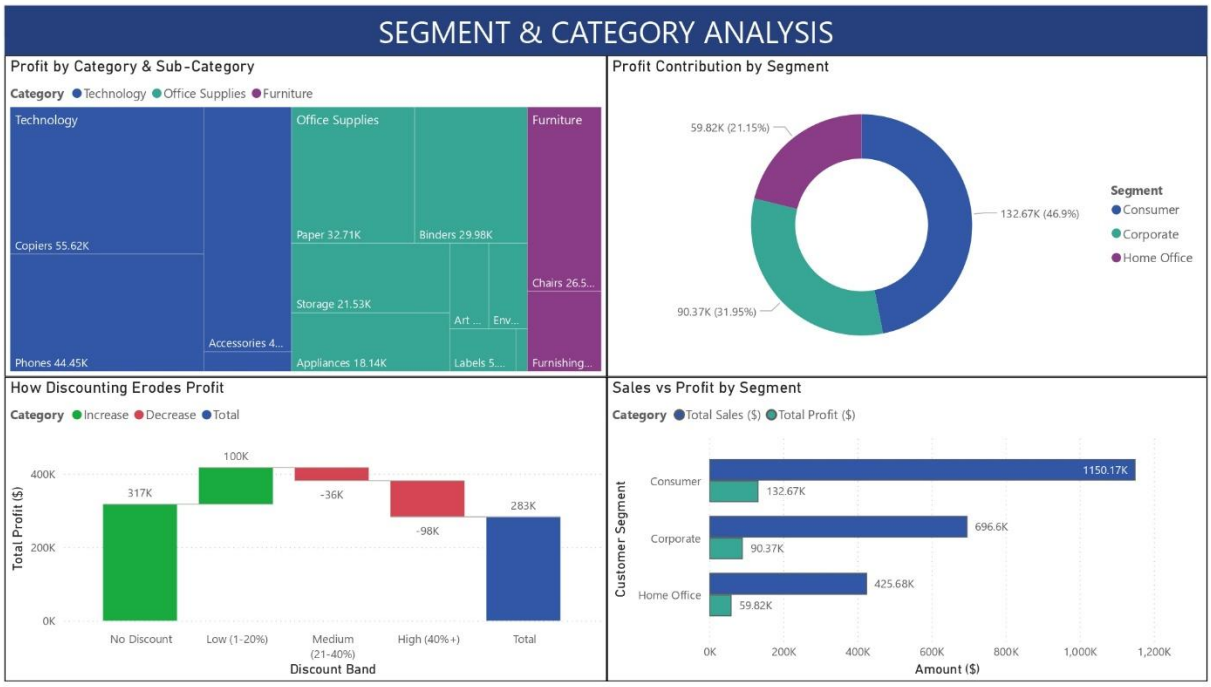
Next Steps:

- **Optimize discount strategy**
  Limit high discount ranges (>20%) that consistently reduce profitability
- **Review loss-making products**
  Adjust pricing or discontinue products that generate sustained losses
- **Focus on high-margin categories**
  Increase emphasis on high-margin categories and sub-categories (e.g., Technology, Labels, Paper)
- **Improve regional performance**
  Investigate operational inefficiencies in the Central region
- **Strengthen pricing governance**
  Implement data-driven pricing strategies aligned with profitability goals

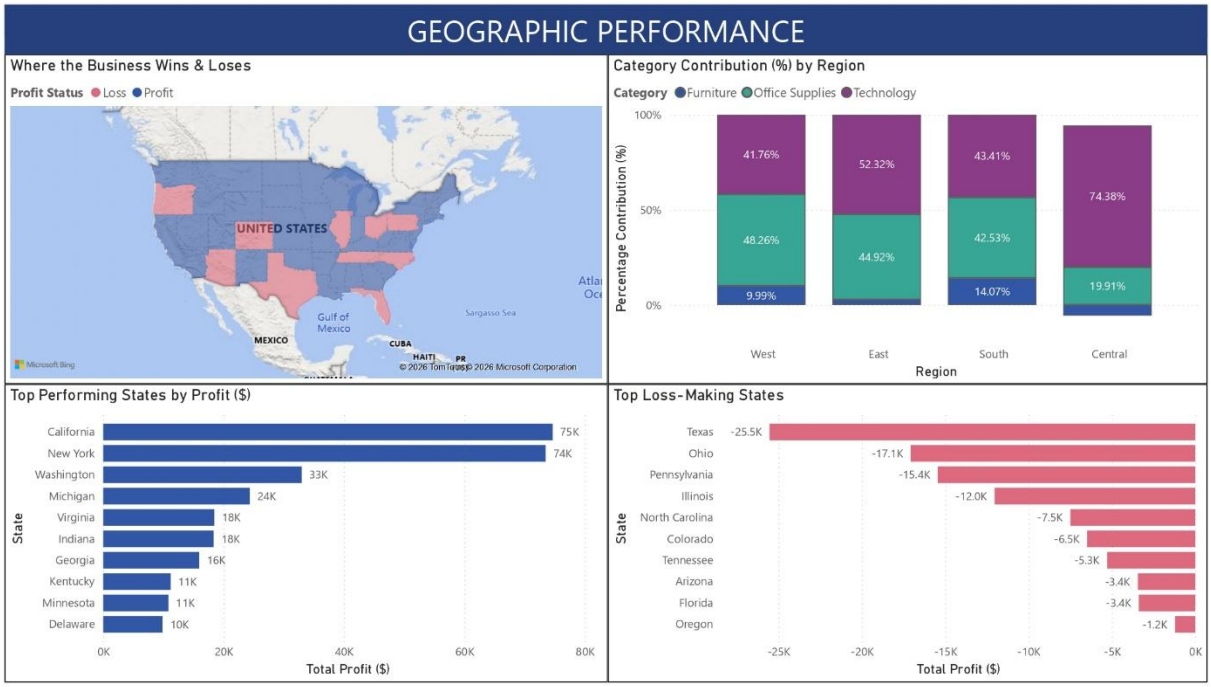
2. Dashboard Preview



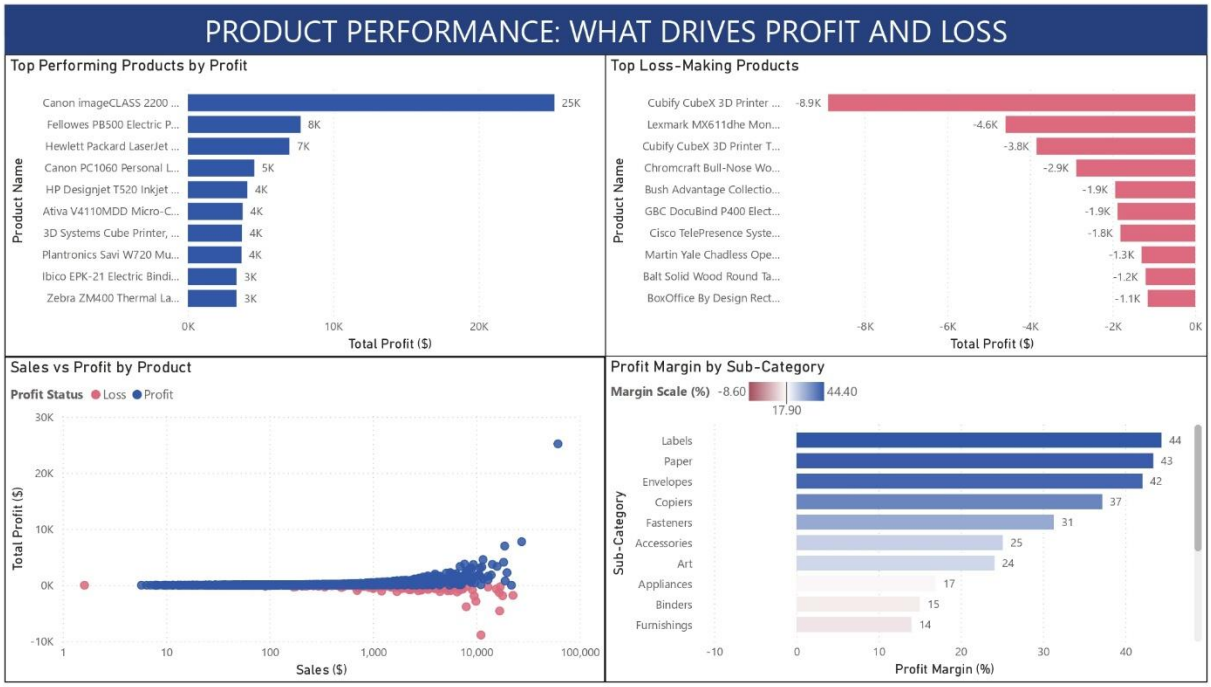
Profitability is uneven across regions, with the Central region underperforming despite overall strong sales performance.



Discounting significantly reduces profitability, with higher discount levels consistently leading to lower margins and losses.



Profit is driven by high-performing states, while losses are concentrated in specific regions and categories.



High sales do not guarantee profitability, as several top-selling products operate at low or negative margins.

3. Business Problem

Retail organizations often prioritize revenue growth without fully understanding profitability drivers. High sales volume does not necessarily translate to strong financial performance, particularly when discounting strategies, product mix, and regional inefficiencies are not optimized.

This analysis aims to evaluate:

- Which regions contribute positively or negatively to overall profitability
- How discounting influences profit margins
- Whether high-selling products generate sustainable profit
- Where operational and pricing inefficiencies exist

4. Methodology

This analysis follows a structured workflow combining SQL-based data preparation, exploratory analysis, and visualization to identify key drivers of profitability and operational inefficiencies.

4.1. Data Cleaning & Transformation (SQL)

- Queried and aggregated transactional data across key dimensions including region, segment, category, state, and product
- Calculated core metrics such as total sales, total profit, and profit margin (%)
- Identified loss-making orders and computed overall loss contribution
- Created structured outputs for integration into Power BI dashboards

4.2. Feature Engineering

- Segmented discount values into bands (No, Low, Medium, High) to evaluate pricing impact
- Derived profitability indicators including:
    - Profit margin (%)
    - Loss vs profit classification
- Generated product-level and regional performance summaries for deeper analysis

4.3. Exploratory Data Analysis (EDA)

- Compared profitability across regions, customer segments, and product categories
- Evaluated the relationship between discount levels and profit outcomes
- Identified concentration of profit and loss across states and products
- Assessed performance variability at both aggregate and granular levels

4.4. Data Modeling & Visualization (Power BI)

- Integrated SQL-derived datasets into a structured Power BI data model
- Created calculated columns and measures using DAX, including:

    - Profit status (profit vs loss)
    - Profit margin (%)
- Designed a 4-page dashboard to present insights across:
    - Overall performance
    - Segment and category drivers
    - Geographic distribution
    - Product-level analysis

4.5. Insight Development

- Synthesized findings across multiple dimensions to identify key profitability drivers
- Connected insights between discounting, product performance, and regional inefficiencies
- Translated analytical findings into actionable business recommendations

5. SQL Analysis & Data Preparation

SQL was used to transform raw transactional data into structured datasets for analysis. Aggregations were performed across multiple dimensions, including region, discount levels, and product performance, to support key business insights.

5.1. Regional Profitability Analysis

```
SELECT
  region,
  ROUND(SUM(sales), 2) AS total_sales,
  ROUND(SUM(profit), 2) AS total_profit,
  ROUND(100 * SUM(profit) / SUM(sales), 1) AS profit_margin_pct
FROM superstore
GROUP BY region
ORDER BY total_profit DESC;
```

- Identified Central as the lowest-performing region, with the weakest profit margin.

5.2. Discount Impact on Profitability

```
SELECT
  CASE
    WHEN discount = 0 THEN 'No Discount'
    WHEN discount BETWEEN 0.01 AND 0.20 THEN 'Low (1-20%)'
    WHEN discount BETWEEN 0.21 AND 0.40 THEN 'Medium (21-40%)'
    ELSE 'High (40%+)'
  END AS discount_band,
  ROUND(SUM(profit), 2) AS total_profit,
```

```
  ROUND(100 * SUM(profit) / SUM(sales), 1) AS profit_margin_pct
FROM superstore
GROUP BY discount_band
ORDER BY profit_margin_pct DESC;
```

- Demonstrated that higher discount levels significantly reduce profit margins.

5.3. Product-Level Profitability

```
SELECT
  `Product Name`,
  ROUND(SUM(sales), 2) AS total_sales,
  ROUND(SUM(profit), 2) AS total_profit
FROM superstore
GROUP BY `Product Name`
ORDER BY total_profit DESC
LIMIT 10;
```

- Identified top-performing and loss-making products, forming the basis of product-level analysis.

5.4. Loss-Making Orders Analysis

```
SELECT
  COUNT(*) AS total_orders,
  SUM(CASE WHEN profit < 0 THEN 1 ELSE 0 END) AS loss_making_orders,
  ROUND(100 * SUM(CASE WHEN profit < 0 THEN 1 ELSE 0 END) / COUNT(*),
1) AS loss_rate_pct
FROM superstore;
```

- Highlighted the proportion of orders contributing to overall losses.

Note: Full SQL scripts used for data preparation and analysis are included in this repository for reference.

6. SKILLS

SQL

- Data aggregation and transformation (SUM, GROUP BY)
- Conditional logic using CASE WHEN
- Creation of derived tables for segmentation and ranking
- Profitability and performance analysis across multiple dimensions

Power BI

- Data modeling and relationship management

- DAX (calculated columns and measures, e.g., profit margin, profit status)
- Dashboard design and layout structuring
- Conditional formatting and KPI visualization
- Advanced visuals (scatter plot, waterfall chart, map, stacked charts)

Data Analysis

- Exploratory Data Analysis (EDA)
- Profitability and margin analysis
- Discount impact analysis
- Geographic and product-level segmentation
- Trend and performance comparison

Analytical Thinking

- Translating data into business insights
- Identifying inefficiencies and root causes
- Connecting insights across multiple dimensions (region, product, pricing)
- Structuring end-to-end analytical narratives

7. Results and Business Recommendations

7.1. Key Insights

- **Profitability varies significantly across regions**
  The West and East regions generate the highest profits, while the Central region underperforms, operating at the lowest margin (8.1%). This highlights regional inefficiencies in performance.
- **Discounting is the primary driver of profit erosion**
  Higher discount levels significantly reduce margins and, in many cases, result in negative profit. This indicates that current discount strategies are not aligned with profitability objectives.
- **High sales do not guarantee profitability**
  Several high-selling products generate low or negative profit, indicating inefficiencies in pricing and discount application.
- **Profit and loss are driven by specific states and products**
  A limited number of states and products contribute disproportionately to overall profit, while losses are concentrated in specific underperforming areas.
- **Profitability varies across sub-categories**
  High-margin performance is driven by select sub-categories (e.g., Labels, Paper), while others such as Furnishings consistently underperform.

7.2. Business Recommendations

- **Optimize discount strategy**
  Limit the use of high discount ranges (>20%) that consistently reduce margins. Implement structured discount thresholds aligned with product-level profitability.

- **Review and rationalize product portfolio**
  Identify and reassess products that generate sustained losses. Consider price adjustments, cost optimization, or discontinuation of consistently underperforming items.
- **Prioritize high-margin categories**
  Increase focus on high-margin categories and sub-categories (e.g., Technology, Labels, Paper) that demonstrate strong profitability.
- **Address regional inefficiencies**
  Investigate the underlying causes of underperformance in the Central region, including pricing strategy, product mix, and operational execution.
- **Implement data-driven pricing strategies**
  Align pricing decisions with profitability metrics rather than sales volume alone. Use data insights to balance competitiveness with margin sustainability.

8. Future Work and Analytical Enhancements

- **Incorporate customer-level analysis**
  Analyze customer lifetime value (LTV), retention, and purchasing behavior to understand how different customer segments contribute to profitability.
- **Develop predictive models for profitability**
  Build models to forecast profit based on pricing, discount levels, and product mix, enabling more proactive decision-making.
- **Perform time-series and seasonality analysis**
  Evaluate monthly and seasonal trends in sales and profit to optimize inventory planning and promotional strategies.
- **Conduct cost and operational analysis**
  Integrate cost-level data (e.g., logistics, supplier costs) to better understand margin variability across regions and product categories.
- **Develop advanced pricing optimization models using predictive analytics**
  Establish data-driven pricing strategies that balance competitiveness with sustainable profitability.
- **Implement automated and real-time dashboards**
  Transition from static reporting to real-time monitoring to track performance and respond quickly to emerging trends.